

Technical Report on Truth Discovery Algorithm

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1 Introduction

This technical report provides comprehensive documentation of our latest truth discovery algorithm, designed to improve the accuracy and reliability of extracting fatality numbers from conflicting claims collected from Twitter (now X). The algorithm utilizes Bayesian updating and a score-based selection scheme to evaluate the data, identifying the most reliable information based on credibility assessments. This approach streamlines the implementation process and demonstrates greater robustness to noisy data compared to the original method proposed in our paper. The report details the implementation steps and performance evaluation, showcasing the algorithm's effectiveness in handling conflicting data sources. For full details on our research, please refer to our paper, "Near-real-time earthquake-induced fatality estimation using crowdsourced data and large-language models", or visit our GitHub repository.

2 Detailed Method

In this section, we provide a detailed introduction to our truth discovery algorithm. The algorithm takes as input the Twitter posts filtered through our event classifier and human loss extractor. The output is the most trustworthy fatality estimates for each time point. The algorithm comprises two main components: a score-based coarse selection and a Bayesian updating-based fine selection of the fatality claims. The first subsection details the score-based selection process, while the second subsection covers the Bayesian-updating-based selection method.

2.1 Score-based Selection

In the Twitter data obtained from the event classifier and human loss extractor, we identify N claims that include a reported fatality number. For each of these claims, we define three scores to assess their truthfulness:

Information Confidence Score ξ : The Information Confidence Score is designed to quantify the confidence level of accurately extracting a human loss value, ranging from 0 to 1.

The score is derived through a process that begins with transforming logits into probabilities using the softmax function, followed by averaging the highest confidence scores for each generated token of the reported death number. Each extracted death number can correspond to multiple tokens, each with a different logit. Initially, the model outputs logits for each potential token, which are unnormalized scores indicating the likelihood of each token being the next in the sequence. These logits are then converted into a probability distribution using the softmax function, ensuring that the probabilities sum to one. The softmax function is defined as:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^V e^{z_j}} \quad \text{for } i = 1, 2, \dots, V$$

where $\mathbf{z} = (z_1, z_2, \dots, z_V)$ represents the logits for a vocabulary of size V , and p_i represents the probability of each token.

Next, for each generated token tok , the model predicts the top- k tokens with the highest probabilities. Let $\text{Tokens}_k(\text{tok})$ denote the set of top- k tokens, and $P_k(\text{tok}) = \{p_{\text{tok}_1}, p_{\text{tok}_2}, \dots, p_{\text{tok}_k}\}$ be their corresponding probabilities. The highest probability among these top- k tokens is selected as the model's most confident prediction:

$$p_{\max}(\text{tok}) = \max(P_k(\text{tok}))$$

The Information Confidence Score, denoted by ξ , is then calculated by averaging these highest probabilities across all tokens in the sequence. The score is computed as:

$$\xi = \frac{1}{|\text{Tokens}|} \sum_{\text{tok} \in \text{Tokens}} p_{\max}(\text{tok})$$

where Tokens is the set of generated tokens in the sequence. This score provides a quantitative measure of the model's overall predictive confidence, reflecting the certainty with which the model predicts the continuation of text sequences.

Information Source Independence Score ρ : The Information Source Independence Score assesses the originality and independence of a tweet text text that indicates casualties. This score ranges from 0 to 1, with higher values suggesting a higher likelihood that the data is original rather than forwarded or copied from other sources. The calculation process involves several key steps, focusing on the detection of duplicate content and normalization to quantify the independence of the information.

First, the text data is preprocessed to remove URLs and irrelevant content, ensuring that the analysis focuses on the meaningful information present in the text. This is achieved by using a regular expression to detect and strip URLs, making the subsequent comparison of texts more accurate.

Next, the dataset is analyzed to identify duplicate content. For each cleaned text, the $\text{Count}(\text{text})$ function calculates how often similar content appears within a defined time slot, such as one hour. This counting step is essential for determining the frequency with which specific information is repeated, which is indicative of how original or independent the content might be.

To emphasize less common and potentially more original content, an "inverted count" $\text{InvertCount}(\text{text})$ is calculated. This step reverses the duplicate counts within the time slot so that more frequently occurring content receives a lower score. The inverted count is computed as follows:

$$\text{InvertCount}(\text{text}) = \text{MaxCount} - \text{Count}(\text{text}) + 1$$

where MaxCount is the maximum count observed within that time slot. The addition of $+1$ prevents the inverted count from being zero, which is crucial for the subsequent normalization step

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to avoid zero values that could lead to null results in multiplicative calculations.

Finally, the inverted counts are normalized to a scale ranging from 0.01 to 1. This normalization ensures that the scores are comparable across different time slots and that all scores fall within a consistent range. The normalization formula is given by:

$$r = 0.99 \times \frac{\text{InvertCount}(\text{text}) - \min(\text{MaxCount})}{\max(\text{MaxCount}) - \min(\text{MaxCount})} + 0.01$$

Here, $\min(\text{MaxCount})$ and $\max(\text{MaxCount})$ are the minimum and maximum inverted counts, respectively, observed in the given time slot of the dataset. The parameters 0.99 and 0.01 ensure that the independence score never reaches exactly zero, thus avoiding complete nullification in any downstream multiplicative processes. This approach guarantees that even the most common content retains a baseline score, preserving the integrity of the scoring system.

Earthquake Fatality Relevance Score r : A binary score, either 0 or 1, that measures the relevance of the text data text regarding casualty information specific to the target disaster event. This score is determined based on the outputs from the event classifier and human loss extractor responses to relevant queries about the event. Both the hierarchical event classifier and the LLM output binary values indicate the relevance and presence of casualty-related information, respectively.

The final relevance score r is defined by the following indicator function:

$$r = \begin{cases} 1 & \text{if } \text{Rel}_{\text{classifier}}(\text{text}) = 1 \text{ and } \text{Rel}_{\text{extractor}}(\text{text}) = 1 \\ 0 & \text{otherwise} \end{cases}$$

Here, $\text{Rel}_{\text{classifier}}(\text{text})$ is a binary value (1 or 0) output by the hierarchical event classifier, indicating whether the text u is relevant to the disaster event. Similarly, $\text{Rel}_{\text{LLM}}(\text{text})$ is a binary value (1 or 0) output by the LLM, indicating the presence of casualty-related information in the text. A score of 1 signifies that the text is relevant to the disaster, while a score of 0 indicates non-relevance. This approach ensures that only texts meeting the criteria of relevance and specific content are flagged as relevant to the disaster event.

User Credibility Score v :

The User Credibility Score is a metric designed to evaluate the credibility and influence of users in the dataset, based on their follower / following ratio and verification status, which is a well-accepted justification for the credibility of tweets.¹⁻³ The calculation begins with the adjustment of follower and following counts by adding 1 to each value, addressing potential issues with division by zero, and ensuring all users are considered. The follower/following ratio is then computed as:

$$\text{Ratio} = \frac{\text{Followers Count} + 1}{\text{Following Count} + 1}$$

Users with a ratio greater than 1 are assigned a score of 1, while those with a ratio of 1 or less receive a score of 0.01. This 0/1 binary variable approach is used to distinguish between more and less credible users, with the 0.01 value applied to avoid nullification in subsequent score calculations. Additionally, the verification status of users is considered, with verified users (denoted by True in the dataset) receiving a score of 1, and non-verified users (denoted by False or missing) receiving a score of 0.01. The final User Credibility Score is calculated by combining these components:

$$v = \text{Ratio} \times \text{Verified}$$

This composite score provides a comprehensive measure of each user's credibility and influence, which is crucial for identifying reliable information sources and improving the quality of data analysis.

Composite Reliability Score τ and selection scheme:

The Composite Reliability Score τ for each data point, which assesses the reliability and relevance of casualty information specific to a disaster event, is determined using the geometric mean of four key components: Information Confidence Score, Information Source Independence Score, Earthquake Fatality Relevance Score, and User Credibility Score. The Composite Reliability Score is calculated using the formula:

$$\tau = \sqrt[4]{\xi \times \rho \times r \times v}$$

The use of the fourth root in this calculation serves to normalize the score, ensuring that it remains within the 0 to 1 range and that contributions from each component are balanced. This normalization helps prevent any single low value from disproportionately lowering the overall score, thereby providing a more even distribution of scores across the dataset.

After computing the Composite Reliability Scores, statistical measures such as the mean and the 99th percentile are calculated to set a threshold for identifying high-quality data points. Data points with a Composite Reliability Score above the 99th percentile are selected (This decision comes from the empirical selection, with an additional criterion excluding entries with reported deaths exceeding a reasonable threshold (maximum number in the PAGER prediction) to avoid including outliers or erroneous data. This threshold is come from experimental result shown in Figure 1

2.2 Bayesian-updating-based Selection

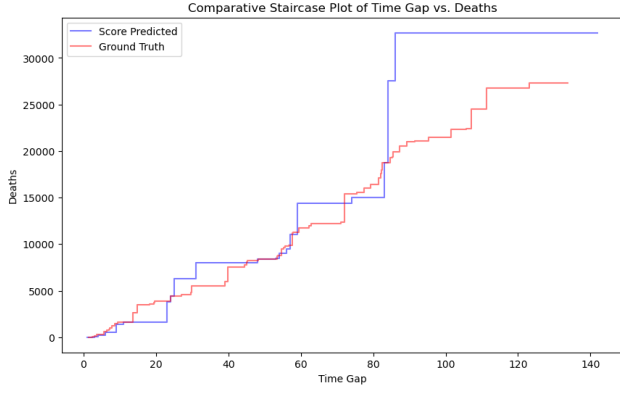
In this section, we outline the methodology for estimating true fatality counts over time using a Bayesian updating approach consistent with the PAGER (Prompt Assessment of Global Earthquakes for Response) system.⁴ The primary goal is to filter out less credible claims and identify the most reliable fatality count for each time point, which then informs the PAGER model for timely and accurate total loss estimation.

The process begins with time-aware categorization of reported fatalities based on the time elapsed since the disaster event. The timestamps of each data point are converted into discrete hourly time slots, calculated as $\lceil T_i - T_{\text{base}} \rceil$, where $\lceil \cdot \rceil$ is the ceiling function, T_i is the actual timestamp for each data point, and T_{base} is the time the earthquake occurred, as documented by USGS. This conversion facilitates chronological analysis, allowing the data to be sorted and analyzed in a time-ordered manner.

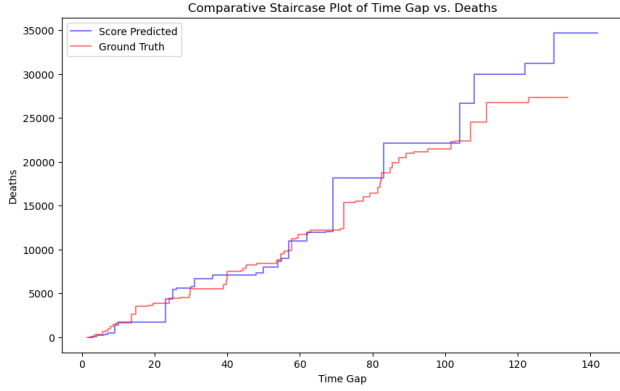
To establish a baseline estimate, the initial prior for the first available time slot is determined using the most frequently reported fatality count, or the mode. This initial value serves as the prior estimate for subsequent Bayesian updates.

In each subsequent time slot, Bayesian updating is applied to refine the prior estimate using new data. The updated estimate provides a new mean μ and standard deviation σ , defining a target range for identifying the true fatality count. Specifically, the target range is set as $[\mu - \sigma, \mu + \sigma]$, allowing for variations within one standard deviation to capture the central tendency of the data.

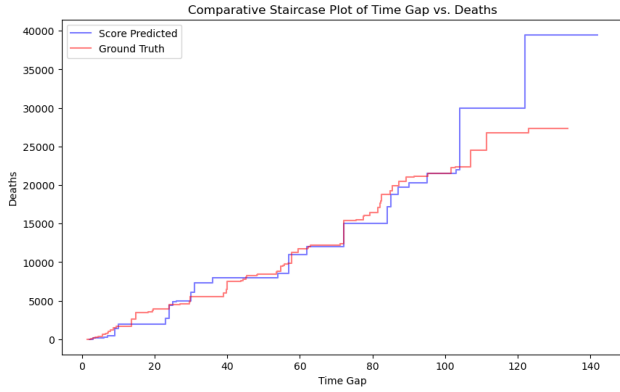
Within this one-sigma range, the actual fatality count closest to the upper boundary of the range is selected as the true estimate.



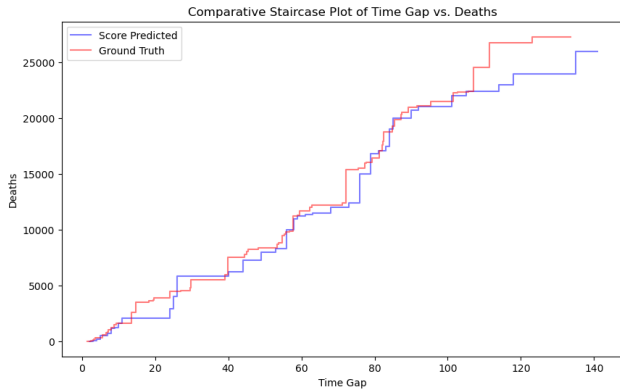
(a) 50th percentile as threshold



(b) 75th percentile as threshold



(c) 85th percentile as threshold



(d) 99th percentile as threshold

Figure 1: Overall comparison of different percentile as the threshold.

To ensure the selected value is within a reasonable range, a constraint is applied: if the selected true estimate exceeds the mode plus three times the standard deviation of the data in the current time slot ($\mu + 3\sigma_{\text{data}}$), the mode itself is chosen. This conservative measure prevents significant overestimations, ensuring the estimates remain realistic and consistent with the most commonly reported figures.

The final true fatality estimates for each time slot are recorded and saved, providing a comprehensive, continuously updated view of the disaster's impact. Additionally, a non-decrease constraint is implemented to ensure that the fatality counts do not decrease over time, maintaining a running maximum of deaths encountered so far. This constraint helps to preserve data integrity, ensuring that the estimates reflect a reasonable progression of events and preventing the data from unrealistically diminishing to zero. This Bayesian updating methodology allows for the systematic integration of new data, continuously refining the accuracy of fatality estimates over time.

2.3 Integrate into the PAGER system

To enhance the PAGER (Prompt Assessment of Global Earthquakes for Response) system's ability to provide timely and accurate fatality estimates, we integrate our refined fatality prediction methodology into the PAGER framework. This integration includes the creation of a staircase plot that displays the total predicted death numbers derived from our filtered data. In this plot, the mean of the final true fatality estimates for each time slot is used as the central estimate, while the variance, representing the data variance before Bayesian updating, indicates the level of uncertainty around these estimates. This visual representation helps to clearly communicate the progression and confidence of fatality predictions over time, aiding decision-makers in their response efforts.

In addition, we overlay a bar plot distribution that shows the fatality estimations from the PAGER system. By presenting both our refined estimates and the PAGER predictions on the same graph, we provide a comprehensive view that allows for direct comparison between the two sets of data. This comparison helps to identify any discrepancies and provides a clearer understanding of the range and distribution of fatality estimates. The integration of these visual elements supports a more informed and coordinated response by highlighting key trends and uncertainties, thereby enhancing the overall effectiveness of the PAGER system in disaster response scenarios.

3 Result

We utilize the last openly available Twitter data from the Turkey earthquake as a case study to demonstrate the effectiveness of our method, presenting results at each step of the process.

Initially, we combined the normalized scores (Information Confidence Score, Information Source Independence Score, and Earthquake Fatality Relevance Score) without incorporating the User Credibility Score. The resulting data revealed that high scores were densely clustered, making it challenging to differentiate between data points, shown in Figure 2. This dense clustering suggests that while the scores are helpful, they alone are insufficient to fully distinguish the reliability of different data points.

Next, we focused on the distribution examining the follower/following ratio and verification status of users, as shown in Figure ?? . The follower/following ratio was categorized into

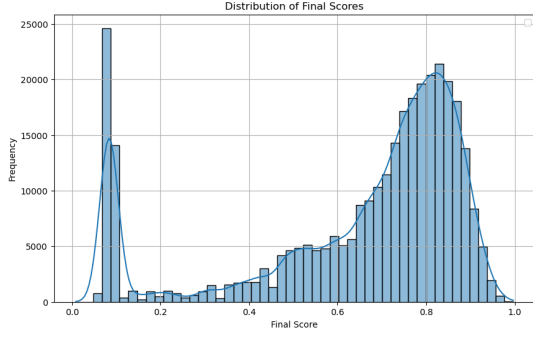
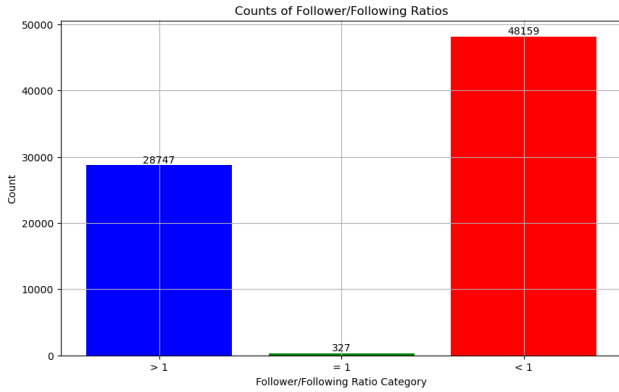
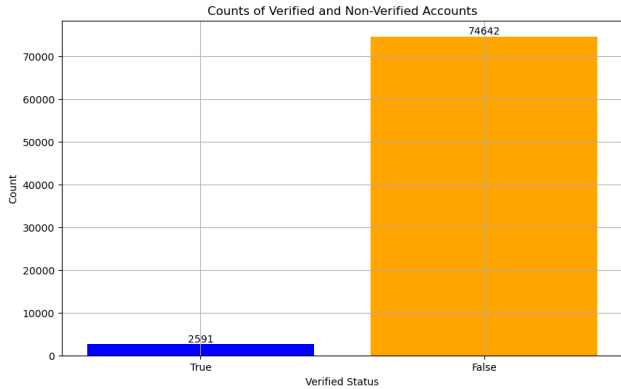


Figure 2: Normalized scores without incorporating the User Credibility Score

three groups: greater than 1, equal to 1, and less than 1. The data revealed distinct gaps between these categories, particularly highlighting differences in credibility. The inclusion of verification status further accentuates these gaps, as verified users generally have higher scores compared to non-verified users. These distinctions are crucial for identifying reliable information sources, as they provide a clear framework for evaluating the trustworthiness of the data.



(a) Follower/following ratio distribution



(b) Verified user distribution

We then combined all components, including the User Credibility Score, to form the Composite Reliability Score, as shown in Figure 4. The distribution of these scores demonstrated a clearer distinction between data points, showing the effectiveness of the User Credibility Score in enhancing the overall assessment of data reliability.

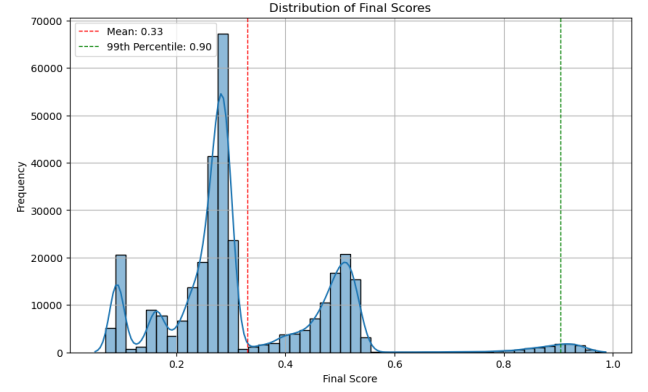


Figure 4: Composite reliability score distribution

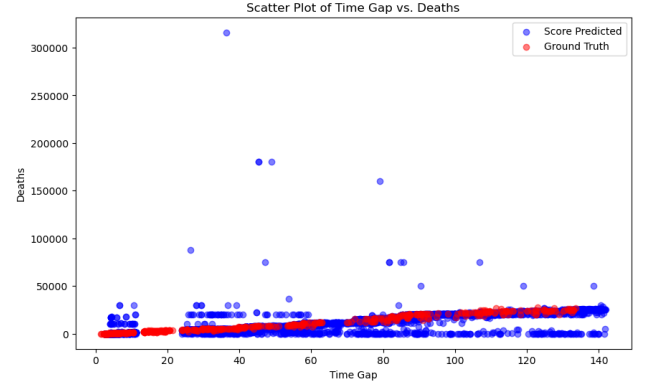


Figure 5: Scatter Plot after the first score-based selection

Following the coarse filtering based on score-based selection, we generated a scatter plot to visualize the data, as shown in Figure 5. Although this filtering step significantly reduced the number of data points, some estimates, historical data, and erroneously extracted information persisted as outliers. These outliers underscore the need for further refinement in the selection process to enhance accuracy.

We then applied Bayesian updating to refine the estimates further, as shown in Figure 6. The resulting scatter plot showed improved consistency and accuracy, with fewer outliers compared to the previous step.

Finally, we employed a staircase plot with a non-decrease physical constraint to depict the cumulative fatality estimates over time, as shown in Figure 7. This plot, which prevents fatality counts from decreasing, aligns closely with the manually selected ground truth data, demonstrating the robustness and accuracy of our approach. The non-decrease constraint ensures that the estimates reflect a realistic progression of events, providing a reliable overview of the disaster's impact.

These results highlight the efficacy of our truth discovery algorithm in processing and analyzing social media data during disaster events, providing valuable insights for timely and accurate disaster response.

4 Conclusion

In this technical report, we have detailed our Dynamic truth discovery algorithm for estimating fatality numbers from conflicting

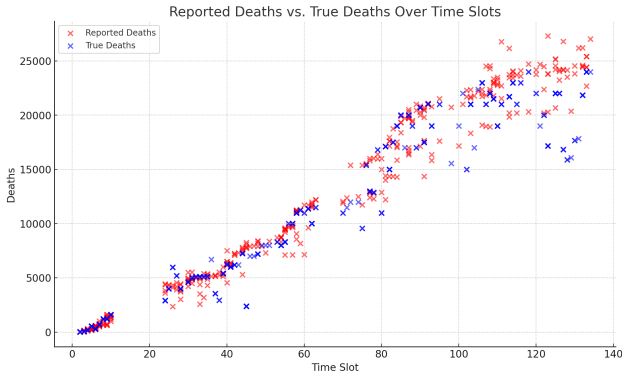


Figure 6: The fatality claim filtered after Bayesian-updating-based selection without non-decrease constraint

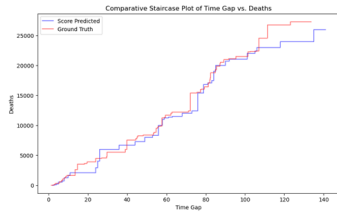


Figure 7: The most credible claims selected compare with ground truth

claims, specifically focusing on data collected from Twitter (now X). The methodology incorporates a Bayesian updating framework, which allows for the systematic refinement of fatality estimates as new data becomes available, ensuring that the predictions are both accurate and reliable. By utilizing a score-based selection process, including Information Confidence Score, Information Source Independence Score, Earthquake Fatality Relevance Score, and User Credibility Score, we are able to assess and prioritize the credibility of each data point. The Composite Reliability Score synthesizes these elements, enabling the identification of the most credible fatality claims. The integration of our refined methodology into the PAGER system provides clear and comprehensive representations of the fatality estimates.

Our approach ensures that the most reliable and accurate data is utilized for disaster response planning and execution. This report showcases the effectiveness of our system in handling noisy and conflicting data, ultimately contributing to better preparedness and response strategies. Future work will aim to further refine these methodologies and explore their applications across various types of disaster scenarios.

5 Limitation

While our dynamic truth discovery algorithm significantly enhances the accuracy and reliability of fatality estimates from social media data, several limitations must be acknowledged. First, the current methodology employs a binary 0/1 scoring system, particularly for the User Credibility Score. This binary approach can oversimplify the complexity of user credibility and influence, potentially leading to less reliable assessments. A more statistical and robust method for evaluating credibility is needed, one that can account for the varying degrees of reliability among users and incorporate additional factors such as user engagement and content

history. Another limitation lies in the empirical data in the PAGER system, which is outdated. The data we relied on may include inaccuracies due to its age. We acknowledge that updating the PAGER data with more recent and accurate information is crucial for improving the precision of our fatality estimates. As newer versions of the PAGER system become available, we plan to integrate these updates to enhance the reliability of our predictions. Lastly, our implementation of Bayesian updating has been simplified compared to the more comprehensive models employed in the original PAGER system. While we focused on a basic form of Bayesian updating, we did not include additional elements such as noise estimation. Future iterations of this work will aim to expand the Bayesian framework to incorporate these complexities, providing a more thorough and accurate representation of disaster impacts. This will involve more sophisticated statistical features that can handle a broader range of variables and uncertainties inherent in disaster response scenarios.

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